# DTFT:

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n} \qquad x[n] = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(\Omega) e^{j\Omega n} \, d\Omega$$

**Ex. 1**

$$x[n] \longrightarrow \overset{\text{LTI}}{\boxed{h[n]}} \longrightarrow y[n]$$

$$y[n] = x[n] * h[n]$$
$$Y(\Omega) = X(\Omega) H(\Omega)$$

Let $x[n] = \alpha^n u[n]$ & $h[n] = \beta^n u[n]$ , $|\alpha| < 1$ , $|\beta| < 1$

$$\Rightarrow X(\Omega) = \frac{1}{1 - \alpha e^{-j\Omega}} \qquad H(\Omega) = \frac{1}{1 - \beta e^{-j\Omega}}$$

$$\Rightarrow Y(\Omega) = \frac{1}{(1 - \alpha e^{-j\Omega})(1 - \beta e^{-j\Omega})} = \frac{A}{1 - \alpha e^{-j\Omega}} + \frac{B}{1 - \beta e^{-j\Omega}}$$

$$= \frac{A(1 - \beta e^{-j\Omega}) + B(1 - \alpha e^{-j\Omega})}{(1 - \alpha e^{-j\Omega})(1 - \beta e^{-j\Omega})}$$

$$\underbrace{A + B = 1 \ \& \ -A\beta - B\alpha = 0}$$

$$\Rightarrow A = \frac{\alpha}{\alpha - \beta} \ , \ B = \frac{-\beta}{\alpha - \beta}$$

$$\Rightarrow Y(\Omega) = \frac{1}{\alpha - \beta} \left[ \frac{\alpha}{1 - \alpha e^{-j\Omega}} - \frac{\beta}{1 - \beta e^{-j\Omega}} \right]$$

$$\Rightarrow y[n] = \frac{1}{\alpha - \beta} \left[ \alpha^{n+1} u[n] - \beta^{n+1} u[n] \right]$$

---

**Ex. 2:**

$H_1(\Omega) \rightarrow$ Freq. Response

$$x[n] \longrightarrow \boxed{h_1[n]} \longrightarrow \boxed{h_2[n]} \longrightarrow x[n]$$

$$h_1[n] * h_2[n] = \delta[n] \iff H_1(\Omega) H_2(\Omega) = 1 \Rightarrow H_2(\Omega) = \frac{1}{H_1(\Omega)}$$

$$h_1[n] = \left(\frac{1}{5}\right)^n u[n] \quad , \quad h_1[n] - A h_1[n-1] = \delta[n]$$

$$A = 1/5$$

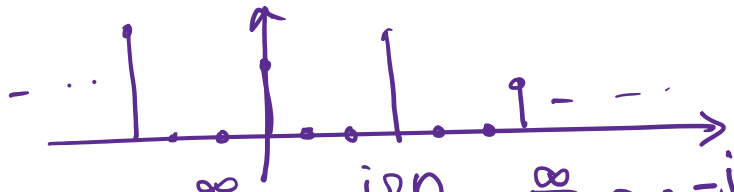
$$H_1(\Omega) = \frac{1}{1 - \left(\frac{1}{5}\right) e^{-j\Omega}} \qquad \Rightarrow h_1[n] * \underbrace{\left(\delta[n] - \frac{1}{5}\delta[n-1]\right)}_{h_2[n]} = \delta[n]$$

$$\Rightarrow H_2(\Omega) = 1 - \left(\frac{1}{5}\right) e^{-j\Omega}$$

$$\Rightarrow h_2[n] = \delta[n] - \frac{1}{5}\delta[n-1]$$

---

**Ex.3** $\quad x_k[n] = \begin{cases} x[n/k] & , n = 0, \pm k, \pm 2k, \ldots \\ 0 & , \text{o.w.} \end{cases}$



} Expans$^n$ in time domain

$$X_k(\Omega) = \sum_{n=-\infty}^{\infty} x_k[n] e^{-j\Omega n} = \sum_{m=-\infty}^{\infty} x[m] e^{-jkm\Omega} = X(k\Omega)$$

} Compres$^n$ in freq. domain.

Signal can't be localized in time & freq. domain simultan. (to arbitrary precis$^n$) } Uncert. Principle

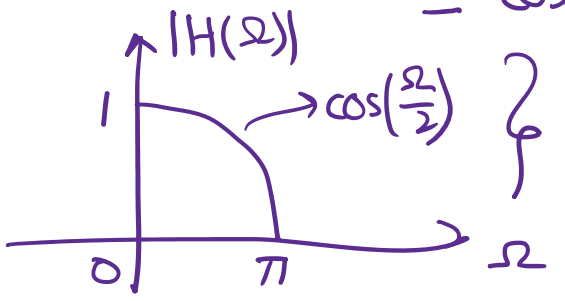
---

**Ex.4** $\quad y[n] = \dfrac{x[n] + x[n-1]}{2}$

$$Y(\Omega) = \frac{X(\Omega) + X(\Omega)e^{-j\Omega}}{2}$$

$$H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{1 + e^{-j\Omega}}{2} = \frac{e^{-j\Omega/2}\left(e^{-j\Omega/2} + e^{j\Omega/2}\right)}{2}$$

$$= \cos\left(\frac{\Omega}{2}\right) e^{-j\Omega/2}$$




} LPF